

Chapter 5: Broken Symmetry and Collective Phenomena:

Exercise #1: (pg 96) "Lagrangian"
(pg 236)

$$S[p, q] = \int_0^t dt \cdot L(q, \dot{q})$$
$$= \int_0^t dt [p \dot{q} - H(p, q)]$$

(3.71) "Field Integral - modified" for units $i/\hbar$

Method #1

$$Z = \int Dq e^{-\frac{i}{\hbar} S[q, \dot{q}]}$$
$$= \int Dq e^{-\frac{i}{\hbar} \int_0^t dt L(q, \dot{q})}$$
$$= \int Dq e^{-\frac{i}{\hbar} \int_0^t dt (p \dot{q} - H(p, q))}$$

Hubbard-Stratonovich transformation

$$e^{-\frac{a}{2} x^2} = \sqrt{\frac{1}{2\pi a}} \int_{-\infty}^{\infty} e^{-\frac{y^2}{2a} - ixy} \cdot dy$$

(3.71) "Field Integral - modified" for units $i/\hbar$

Method #2

$$\begin{aligned}
 Z &= \int Dq e^{-\frac{i}{\hbar} S[q, \dot{q}]} \\
 &= \int Dq e^{-\frac{i}{\hbar} \int_0^t L(q, \dot{q}) dt} \\
 &= \int Dq e^{-\frac{i}{\hbar} \int_0^t \left(\frac{1}{2} m \dot{q}^2 - V(q) \right) dt}
 \end{aligned}$$

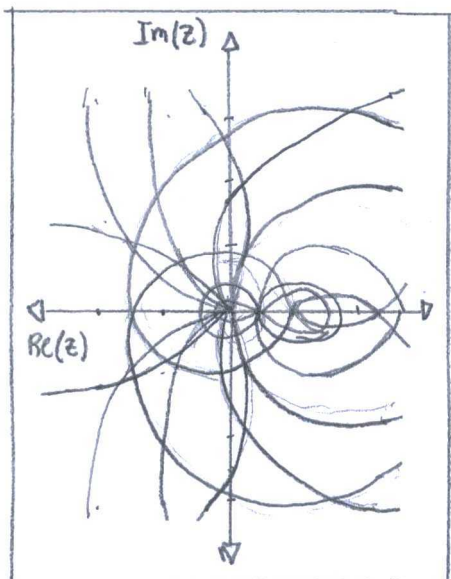
Kinetic energy
Potential energy

$$= \iint Dq Dp e^{-\frac{i}{\hbar} \int_0^t \left(\frac{1}{2} p \cdot \dot{q} - H(x, p) \right) dt}$$

... When $p = m \dot{q}$

$$\text{kg} \cdot \frac{\text{m}}{\text{s}} = \text{kg} \cdot \frac{\text{m}}{\text{s}}$$

Exercise #2:
(pg 244)



Cite: Riemann, B.

"Ueber die Anzahl der Primzahlen unter einer gegebenen Grösse"

Berliner Akademie (1859)

"Reimann Zeta Function"

$$\Gamma(s-1)\zeta(s) = \int_0^{\infty} \frac{x^{s-1}}{e^x - 1} dx$$

The public shifted the gamma for solutions at one:

$$\zeta(s) = \frac{1}{\Gamma(s)} \int_0^{\infty} \frac{x^{s-1}}{e^x - 1} dx$$

... useful to the exercise.

(from problem) "Three-dimensional
free particle spectrum"

$$E = \hbar^2 k^2 / 2m$$

(from problem) "Density of a
condensate"

$$\rho_0(T) = \rho [1 - (T/T_c)^{3/2}]$$

(from problem) "Average interparticle
spacing"

$$a = \rho^{-1/3}$$

Equation of state:

$$P \cdot V = n R T$$

$$\frac{n}{V} = \frac{P}{R \cdot T}$$

$$= \frac{(2s+1)}{(2\pi\hbar)^3} \frac{1}{RT} \int_0^\infty \frac{dp_x dp_y dp_z}{e^{\beta(\epsilon - \mu)} - 1}$$

$$= \frac{(2s+1) m^{3/2}}{2 \cdot \pi^2 \cdot \hbar^3} \int_0^\infty \frac{p^2 dp}{e^{\beta(\epsilon - \mu)} - 1} \dots$$

$$= \frac{(2s+1) m^{3/2}}{2^{1/2} \cdot \pi^2 \cdot \hbar^3} \int_0^\infty \frac{\sqrt{\epsilon} d\epsilon}{e^{\beta(\epsilon - \mu)} - 1} \dots \epsilon = \frac{p^2}{2m}$$

$$= \frac{(2s+1) (k_B T)^{3/2} m^{3/2}}{2^{1/2} \cdot \pi^2 \cdot \hbar^3} \int_0^\infty \frac{\sqrt{\epsilon}^{1/2} d\epsilon}{e^{-\beta\mu} \cdot e^{\epsilon/T} - 1}$$

$$= \frac{(2s+1) (k_B T)^{3/2} m^{3/2}}{2^{1/2} \cdot \pi^2 \cdot \hbar^3} \cdot T^{(3/2)} \cdot \zeta(3/2) \dots T^{(3/2)} = \frac{\sqrt{\pi}}{2}$$

$$= (2s+1) \cdot \left(\frac{m k_B T}{2 \pi \hbar^2} \right)^{3/2} \zeta(3/2)$$

"Thermal wavelength"

$$= P$$

$$T_c = \frac{2 \pi \hbar^2}{m k_B a^2 \zeta(3/2)}$$

$$= \lambda^{-d+2} \int d^d k^n e^{-i k^n \cdot r} \cdot G(k^n)$$

$$= \lambda^{-d+2} \cdot G(r)$$

Where $G(r) = \frac{1}{p^2 + m^2}$

If $d > 2$: Fourier Transform has
a definition.

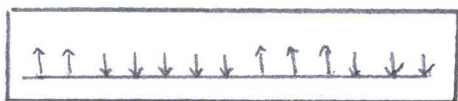
$d = 2$: Fourier transform has
no definition.

$d < 2$: Exponential decay
and 'long range'
bias appear.

Exercise #4:
(pg 253)

(from problem) "Partition function"

$$Z = \sum_{\{S\}} \exp(-\beta H)$$



"Spin configurations"

(pg 253) "one-dimensional Ising
Chain"

$$H = -J \sum_{i=1}^N S_i \cdot S_{i+1} - B \sum_{i=1}^N S_i$$

$$= \frac{C_0 \hbar^2}{m \cdot a^2} \quad \dots \text{where } C_0 = \frac{2\pi}{k_B \zeta(3/2)}$$

Density of particles in condensate ($T < T_c$):

$$\rho = \rho_0 + \rho_c$$

$$\rho_0 = \rho - \rho_c$$

$$= \rho [1 - \rho_0 / \rho]$$

$$= \rho \left[1 - (T/T_c)^{3/2} \right]$$

Exercise #3:
(pg 250)

(from problem) "Propagator"

$$G(r \equiv |r|) = \langle \phi(r) \phi(0) \rangle$$

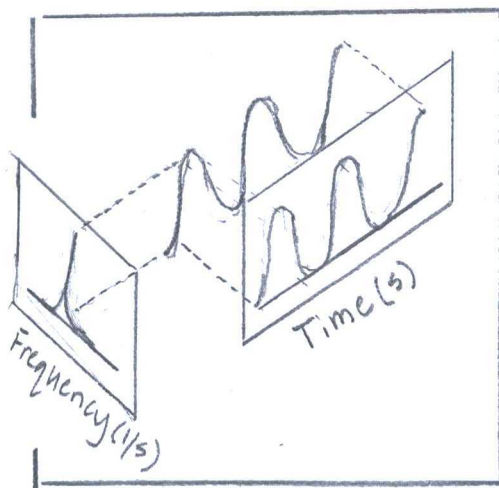
() "Fourier Transform"

$$G(r) = \int dr e^{-ikr} F(r)$$

$$G(\lambda r) = \int d^d k e^{-ik \lambda r} \cdot G(k)$$

$$= \int d^d(k/\lambda) e^{-ik^n \cdot r} \cdot G(k^n/\lambda) \dots k^n = \lambda k$$

$$= -\lambda^{-d} \int d^d k^n \cdot \lambda^2 \cdot G(k^n)$$



"Fourier
Transform"

Where $S_i \in \{1, -1\}$

$$= -J(\# \text{ Particles} - \# \text{ Particles opposite spin})$$

$$= -B(\# \text{ Particles opposite spin})$$

$$= -J(N - k) - B(k)$$

"one particle"

$$Z = \sum_{\text{enum}} \exp(-\beta[-J(N-k) - B(k)])$$

$$= \sum_{\text{enum}} \frac{N!}{k!(N-k)!} \exp(\beta[J(N-k) + B(k)])$$

$$= \sum_{\text{enum}} \binom{N}{k} \exp(\beta[J(N-k) + B(k)])$$

Temperature dependence:

$$\lim_{T \rightarrow 0} Z = \lim_{T \rightarrow 0} \sum_{\text{enum}} \frac{N!}{k(N-k)!} e^{\frac{1}{k_B T} [J(N-k) + B(k)]}$$

$= \infty$ "No ordering at low-temperatures"

Exercise #5: (5.13) "Effective action by Hamilton"

(pg 256)

$$S[\rho, \phi] = \int d\tau \int d^d r \left(i \delta \rho \partial_\tau \phi + \frac{\rho_0}{2m} (\nabla \phi)^2 + \frac{g \rho^2}{2} \right)$$

A mass coefficient implies imaginary quantities, a ^{new} coefficient for units, and roots without mass in the equation for motion.

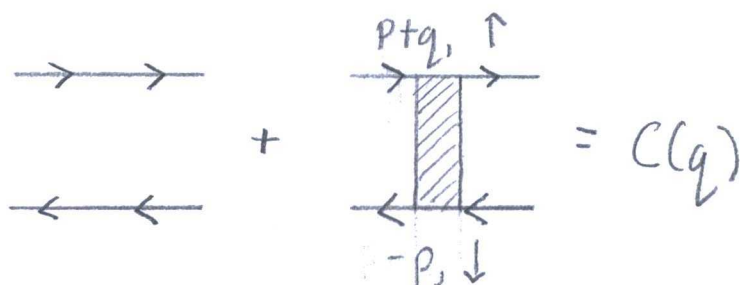
(5.13) "Effective action by Lagrangian"

$$S[\phi] \approx \frac{1}{2} \int d\tau \int d^d r \left(\frac{1}{g} (\partial_\tau \phi)^2 + \frac{\rho}{m} (\nabla \phi)^2 \right)$$

A mass coefficient implies roots with no real mass. An equation of motion needs mass.

Exercise #6:

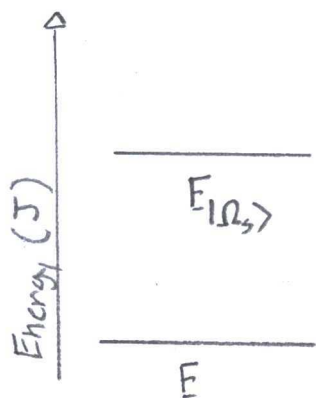
(pg 260)



$$= \left\langle \frac{T}{L^d} \sum_k \bar{\psi}_{k+q, \uparrow} \psi_{-k, \downarrow} \right. \\ \left. \times \frac{T}{L^d} \sum_{k'} \bar{\psi}_{k'+q, \downarrow} \psi_{-k', \uparrow} \right\rangle$$

Exercise #7:
(pg 265)

(pg 265) "Total Energy in Ground
State for Fermi Sea"



$$E_{|\Omega_s\rangle} \equiv \langle \Omega_s | \hat{H} - \mu \hat{N} | \Omega_s \rangle$$

$$= \sum_k (\xi_k - \lambda_k) + \Delta^2 L^d / g$$

$$E_0 = \lim_{\Delta \rightarrow 0} E_{|\Omega_s\rangle}$$

$$= \lim_{\Delta \rightarrow 0} \sum_k (\xi_k - \lambda_k) + \Delta^2 L^d / g$$

$$= \sum_k (\xi_k - \lambda_k) < E_{|\Omega_s\rangle}$$

(5.26) "Input Parameter"

$$\Delta = \frac{\omega_D}{\sinh(1/gv)} \stackrel{gv \ll 1}{\simeq} 2\omega_D \exp\left(\frac{-1}{gv}\right)$$

$$\lim_{\Delta \rightarrow 0} \partial_{\Delta} E_{|\Omega_s\rangle} = \lim_{\Delta \rightarrow 0} \Delta$$

$$= 0$$

Exercise #8: (pg 266) "Fermi Surface"
(pg 267)

$$\rho(\epsilon) = \frac{1}{L^d} \sum_{k\sigma} \delta(\epsilon - \lambda_k)$$

$$= \int d\xi \frac{1}{Ld} \sum_{k\sigma} \delta(\xi - \xi_k) \delta(E - \lambda(\xi))$$

$$\cong V \sum_{s=\pm 1} \int_0^\infty d\xi \frac{\delta(\xi - s[E^2 - \Delta^2]^{1/2})}{|2\xi| \xi^2 + \Delta^2|^{1/2}}$$

$$= 2V \Theta(E - \Delta) \frac{E}{(E^2 - \Delta^2)^{1/2}}$$

$$\int_\Delta^E dE \rho^{(E)} = \int_\Delta^E dE \cdot 2V \Theta(E - \Delta) \frac{E}{(E^2 - \Delta^2)^{1/2}}$$

$$= \frac{2V}{2} \int_\Delta^E \frac{1}{\sqrt{u}}$$

$$= 2V \sqrt{E^2 - \Delta^2}$$

$$\simeq 2VE \text{ , when } E \gg \Delta$$

<u>u-substitution</u> $u = E^2 - \Delta^2$ $\frac{du}{dE} = 2E$

Exercise #9:
(pg 267)

(from problem) "Lattice d-wave
Superconductor"

$$\Delta_k = \Delta_0 (\cos(k_x a) - \cos(k_y a))$$

$$\text{where } k = (k_x, k_y)^T$$

@ $k_x = \pm k_y$:

(From problem) "Kinetic Energy"

$$t_k = -t(\cos(k_x a) + \cos(k_y a))$$

(from problem) "Bogoliubov Hamiltonian"

$$\begin{aligned}\hat{H} &= \sum_k \psi_k^\dagger ((t_k - \mu)\sigma_3 + \Delta_k \sigma_1) \psi_k \\ &= \sum_k \psi_k^\dagger ((-t(\cos(k_x a) + \cos(k_y a)) - \mu)\sigma_3 \\ &\quad + \Delta_0 (\cos(k_x a) - \cos(k_y a))\sigma_1) \psi_k\end{aligned}$$

$$\begin{aligned}\hat{H} &= \sum_k \psi_k^\dagger ((-t(\cos(\frac{\pm\pi}{2a}a) + \cos(\frac{\pm\pi}{2a}a)) - \mu)\sigma_3 \\ &\quad + \Delta_0 (\cos(\frac{\pm\pi}{2a}a) - \cos(\frac{\pm\pi}{2a}a))\sigma_1) \psi_k\end{aligned}$$

... from exercise $k = (\pm 1, \pm 1)\pi/2a$

$$= \sum_k \psi_k^\dagger (-\mu \sigma_3) \psi_k$$

= 0 ... When chemical potential (μ)
is zero.

Matrix Formation

$$\hat{H}_{++} = \begin{pmatrix} \epsilon a(k_x + k_y) & \Delta_0 a(k_x - k_y) \\ \Delta_0 a(k_x - k_y) & -\epsilon a(k_x + k_y) \end{pmatrix}$$

scaled
 $\xrightarrow{k_1 \rightarrow \frac{\Delta_0}{\epsilon} k_1}$ $-k_1 - k_2$

$$= k_1 \sigma_1 + k_2 \sigma_2$$

• With "Pauli matrices"

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

Density of States:

(pg 266) "Density of Excitations"

$$\rho(\epsilon) = \frac{1}{L^d} \sum_{k,d} \delta(\epsilon - \lambda_k)$$

$$= \int d\xi \frac{1}{L^d} \sum_{k,\sigma} \delta(\xi - \xi_k) \cdot \delta(\epsilon - \lambda(\xi))$$

$$= \int d\xi \delta(\epsilon - \lambda(\xi))$$

$$\sim |\epsilon|$$

Exercise #10: (5.19) "a term in Bethe-Salpeter"

(pg 272)

$$\frac{T}{L^d} \sum_p G_p \cdot G_{-p} = \int_{-W_D}^{W_D} d\epsilon V(\epsilon) \frac{1 - 2n_F(\epsilon)}{2G}$$

$$\simeq V \int_T^{W_D} \frac{d\epsilon}{\epsilon}$$

$$= V \ln \left(\frac{W_D}{T} \right)$$

(from problem) "Fermi expansion"

$$n_F(\epsilon, T) - n_F(\epsilon, T_c) \simeq (T - T_c) \partial_T \Big|_{T_c} n_F(\epsilon, T)$$

$$= -\partial_c n_F(\epsilon, T_c) (T - T_c) \frac{\epsilon}{T_c}$$

Exercise #11:

(pg 286)

Current-field relation:

(pg 285) "External Electric field"

$$E = -i 2\pi A \quad \leftarrow \text{Not in my E\&M book.}$$

(5.49) "Current density"

$$\begin{aligned} \vec{j}(r) &= \frac{\delta}{\delta A(r)} \int d^d r \cdot \frac{n_s}{2m} A^2 \\ &= \frac{n_s}{m} A(r) \end{aligned}$$

(from problem) "Newton's equation of motion"

$$m \ddot{r} = E \quad \leftarrow \text{modified Lorentz' Law, not}$$

$$\dot{r} = \frac{E}{m} d\tau$$

From problem with
uneven units.

$$\vec{j}(r) = n \cdot e \cdot \vec{v}$$

$$= n \cdot e \cdot \dot{r}$$

$$= n \cdot e \cdot \frac{E}{m} d\tau$$

$$\partial_\tau \vec{j}(r) = \frac{n_s}{m} e E$$

Current-field relation with friction:

(from problem) "with friction"

$$m \ddot{r} = -\frac{m}{\tau} \dot{r} + E$$

$$\dot{r} = -\frac{1}{\tau} r + \frac{E}{m} \tau$$

... friction and time are the same

$$2\pi j(r) = n_s \left(\frac{E}{m} - \frac{\dot{r}}{\tau} \right)$$

symbol (τ) by author. Maybe friction in time!

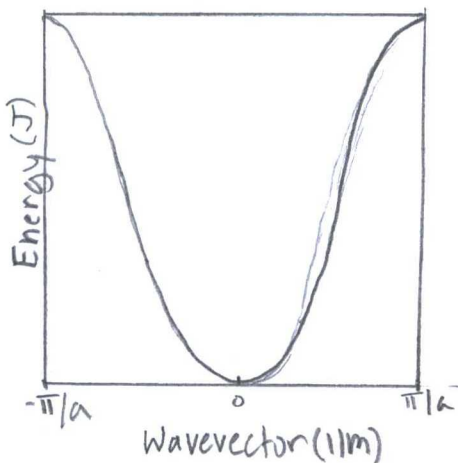
Exercise #12:
(pg 305)

Problem 5.6.1 - Peierls Instability:



a) (from problem) "Action function for an electron"

$$S_{el}[\psi] = \int d\tau \int dx \bar{\psi} \left(\partial_\tau - \frac{\partial_x^2}{2m} - \mu \right) \psi$$



(from problem) "Action function electron-phonon"

$$S_{el-ph}[\psi] = g \int d\tau \int dx \bar{\psi} \psi \partial_x u$$

(from problem) "Full Euclidean action"

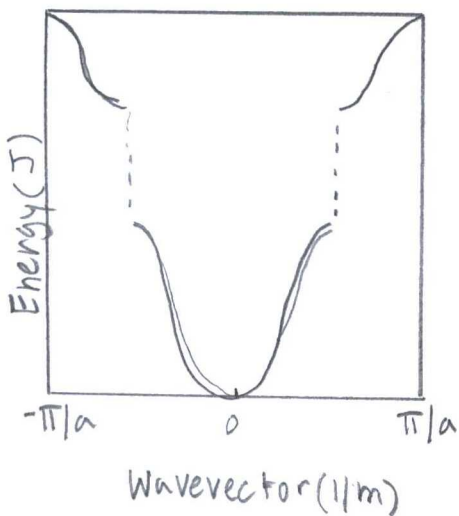
$$S = S_{el} + S_{ph} + S_{el-ph}$$

() "Field Integral"

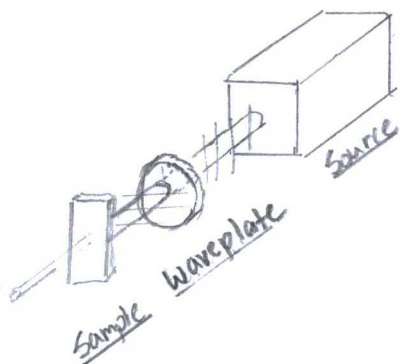
$$Z = \int D\psi e^{-S[\psi]}$$

$$= \int D\psi e^{-S_{el} - S_{el-ph}}$$

$$= \int_{-\infty}^{\infty} D\psi e^{-\int_0^\tau d\tau \int_{-\infty}^{\infty} dx \bar{\psi} \left(\partial_\tau - \frac{\partial_x^2}{2m} - \mu \right) \psi - g \int_0^\tau d\tau \int_{-\infty}^{\infty} dx \bar{\psi} \psi \partial_x u}$$



I struggled on the bounds and solution to the integral. The answer is correct when $\int_{-\infty}^{\infty} dx = \int_{-\infty}^{\infty} \delta(x) dx = 1$, Dirac Delta.



"Peierls Instability"

$$= \sqrt{\frac{\pi}{\left(\partial_\tau - \frac{\partial_x^2}{2m} - \mu - g \partial_x u \right) \tau}}$$

$$\begin{aligned}
&= \sqrt{\pi} e^{-\frac{1}{2} \left[\ln \left(\partial_\tau^2 - \frac{\partial_x^2}{2m} - \mu + g \partial_x u \right) \tau \right]} \\
&= \sqrt{\pi} e^{-\frac{1}{2} \ln [G_0^{-1} + g(\partial_x u)]} \\
&= \sqrt{\pi} e^{-\frac{1}{2} \ln \left[G_0^{-1} - g \left(\frac{g}{2} \sum_{q, q'} q^2 \chi(q, \omega_m) |u_{q, q'}|^2 \right) \right]}
\end{aligned}$$

$$S_{\text{eff}}[u] = S_{\text{ph}}[u] + \Delta S_{\text{el, el-ph}}$$

(from problem) "Action function phonon"

$$S_{\text{ph}}[u] = \frac{\rho}{2} \int d\tau \int dx \left((\partial_\tau u)^2 + c^2 (\partial_x u)^2 \right)$$

$$S_{\text{eff}}[u] = \left(\frac{\rho}{2} \sum_{\tau} \sum_x (\omega_m^2 + c^2 q^2) - \frac{g^2}{2} \sum_{q, q'} q^2 \chi(q, \omega_m) |u_{q, q'}|^2 \right)$$

b) (from exercise) "Susceptibility"

$$\chi(q, \omega_m) = -T L^{-1} \sum_{p, \omega_n} G_{0, p, \omega_n} G_{0, p+q, \omega_n + \omega_m}$$

At $\omega_m = 0$ and $u = \pi u_0$,

$$\begin{aligned}
S_{\text{eff}}[u] &= \left(\frac{\rho}{2} \sum_{\tau} \sum_x c^2 q^2 + \frac{g^2}{2} \sum_{q, q'} q^2 T L^{-1} \sum_{p, \omega_n} \right. \\
&\quad \left. G_{0, p, \omega_n} \circ G_{0, p+q, \omega_n} \right) |u_{q, q'}|^2
\end{aligned}$$

$$\cong \frac{\pi^2 u_0^2}{2} \left[c^2 q^2 + \frac{g^2}{2} \cdot q^2 \cdot T \cdot L^{-1} \right]$$

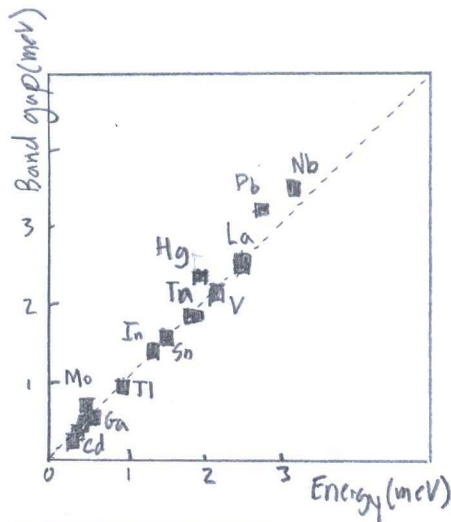
$$\text{At } S_{\text{eff}}[u] = 0,$$

$$S_{\text{eff}}[u] = \pi^2 u_0^2 [pc^2 + g^2 \cdot T]$$

$$= 0$$

$$T_c = \frac{pc^2}{g}$$

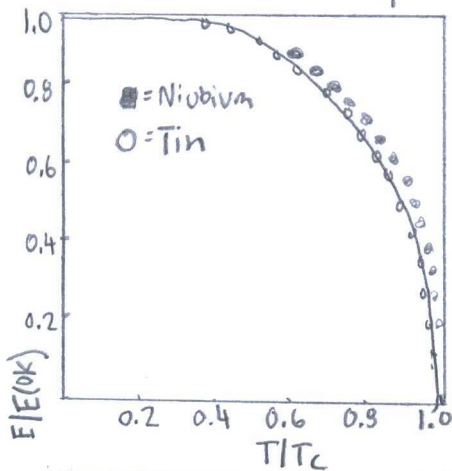
Problem 5.6.2 - Temperature Profile of the BCS gap:



(5.33) BCS Bandgap Equation

$$\frac{1}{g\nu} = \int_0^{W_0/2T} dx \frac{\tanh(x^2 + k^2)^{1/2}}{(x^2 + k^2)^{1/2}}$$

where $k = \Delta/2T$



$$a) \approx \int_0^{W_0/2T} \frac{1}{2} \frac{\tanh(\xi)^{1/2}}{\xi} d\xi \quad \text{... when } \xi = x^2 + k^2$$

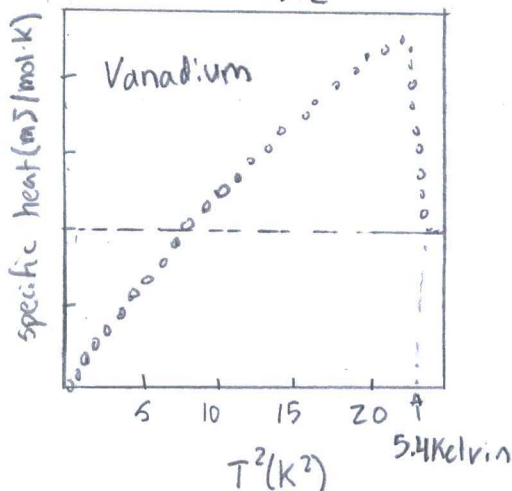
$$\approx \frac{1}{2} \int_1^{W_0^2/2T} \frac{1}{\xi} d\xi \quad \text{... because } \tanh(\xi)^{1/2} \sim 1 \text{ beyond } \xi > 1$$

$$\approx \ln(W_0/2T)$$

$$T_c = \frac{W_0}{2} e^{-1/g\nu}$$

b) (5.34) "Temperature shift"

$$\Delta_0 = \text{const} \times \sqrt{T_c(T_c - T)}$$



Temperatures close to T_c :

$$\frac{1}{g\nu} - \frac{\Delta T}{T_c} = - \int_0^{W_D^2/2T} d\xi \left(\frac{\tanh(\xi + K^2)^{1/2}}{(\xi + K^2)^{1/2}} - \frac{\tanh(\xi)}{\xi} \right) + \frac{1}{2} \int_0^{W_D^2/2T} d\xi \frac{\tanh \xi}{\xi}$$

$$-\frac{\Delta T}{T_c} = \frac{1}{2} \int_0^{W_D^2/2T} d\xi \left(\frac{\tanh(\xi + K^2)^{1/2}}{(\xi + K^2)^{1/2}} - \frac{\tanh \xi}{\xi} \right)$$

If $0 \leq \xi \leq 1$, then

$$-\frac{\Delta T}{T_c} \simeq \frac{K^2 \cdot W_D}{2T}$$

$$\simeq \text{const} \cdot K^2$$

$$\simeq \text{const} \cdot \frac{\Delta^2}{T_c^2}$$

$$\Delta_0 \simeq \text{const} \cdot \sqrt{T_c(T_c - T)} \quad \text{as when } \delta T = T - T_c$$

If $1 \leq \xi \leq W_D^2/4T^2$, then

$$-\frac{\Delta T}{T_c} \simeq K^2$$

Problem 5.6.3 - Fluctuation contribution to the superconductor action:

(5.36) "Quadratic action"

$$S^{(2)}[\Delta] = \sum_q T_q^{-1} |\Delta(q)|^2, \quad T_q^{-1} = \frac{1}{g} - \frac{T}{L^d} \sum_p G_p G_{-p+q}$$

(From problem) "Kernel"

$$\chi_q^c \equiv \frac{-T}{L^d} \sum_p G_p G_{-p+q}$$

$$= -\frac{1}{L^d} \sum_p \frac{-n_F(\xi)}{i\omega - \xi_p - \xi_{p_2}} + \frac{-n_F(\xi - i\omega_m)}{i\omega + \xi_p - \xi_{p_2}}$$

$$\approx -\frac{1}{L^d} \sum_p \frac{[n_F(\xi_p) - n_F(\xi_{q-p})]}{i\omega - \xi_p - \xi_{-p+q}}$$

• from problem 3.8.8.

$$= \prod_m \int dV e^{\frac{T(1-e^{-i\omega\tau})}{\omega_0 V}} \frac{1}{\beta Z} \sqrt{\frac{1}{4\pi E_c}} \int dV e^{-S[V]} Z[V] G_\alpha[V](\tau)$$

c) (from problem) "Identity"

$$\sum_{k=1}^{\infty} \frac{\cos(kX)}{k^2} = \frac{\pi^2}{6} - \frac{\pi|X|}{2} + \frac{X^2}{4} + \dots$$

$$G(\tau) = \prod_m \int dV e^{\frac{T(1-e^{-i\omega\tau})}{\omega_0 V}} \frac{1}{\beta Z} \sqrt{\frac{1}{4\pi E_c}} \int dV e^{-S[V]} Z[V] G_\alpha[V](\tau)$$

$$= F(\tau) \cdot G_\alpha(\tau)$$

$$\text{where } F(\tau) = \prod_m \int dV e^{\frac{T(1-e^{-i\omega\tau})}{\omega_0 V}} \frac{1}{\beta Z} \sqrt{\frac{1}{4\pi E_c}}$$

$$G_\alpha(\tau) = \int dV e^{-S[V]} \cdot Z[V] \cdot G_\alpha[V](\tau)$$

d) Bifurcations:

$$\text{Roots: } \frac{\partial}{\partial V} \left(\frac{V^2}{4E_c} - i(N_0 V + \bar{\psi}_\alpha V \psi_\alpha) \right)$$

$$= 0, \quad V = 2iE_c(N_0 - \bar{\psi}_\alpha \psi_\alpha)$$

$$\text{Equation: } G(\tau) = F(\tau) \cdot G_{\alpha_0}(\tau)$$

$$= F(\tau) \cdot \iint DV D\psi e^{-\int d\tau \sum_\alpha (\partial_\tau + E_\alpha - \mu) - \frac{2iE_c(N_0 - \bar{\psi}_\alpha \psi_\alpha)}{4E_c}}$$

$$\hookrightarrow -i(N_0(N_0 - \bar{\psi}_\alpha \psi_\alpha) + \bar{\psi}_\alpha(N_0 - \bar{\psi}_\alpha \psi_\alpha))$$

$$= F(\tau) \int D\psi DV e^{-\int d\tau (\sum_k (2\tau + \epsilon_k - \mu) - E_0 (N_0 - \sum_k \psi_k^\dagger \psi_k)^2)}$$

e) (from problem) "Function"

$$F(\tau) = \exp(-E_c(\tau - \frac{1}{\beta\tau^2}))$$

$$\simeq \exp(-E_c \cdot \tau) \quad \dots \text{when } E \gg T_c$$

Fourier Transform:

$$\begin{aligned} F &= \int_0^\beta e^{-i\omega\tau} \cdot F(\tau) \cdot d\tau \\ &= \int_0^\infty (e^{i\omega\tau} + e^{-i\omega\tau}) e^{-E_c\tau} d\tau \\ &= \int_0^\infty 2 \cdot \cos(\omega\tau) \cdot e^{-E_c\tau} d\tau \\ &= \frac{2E_c}{E_c^2 + \omega^2} \end{aligned}$$

Laplace Transform

$$\begin{aligned} \int_0^\infty e^{-\alpha\tau} \cos(\omega\tau) d\tau \\ = \frac{s + \alpha}{(s + \alpha)^2 + \omega^2} \end{aligned}$$

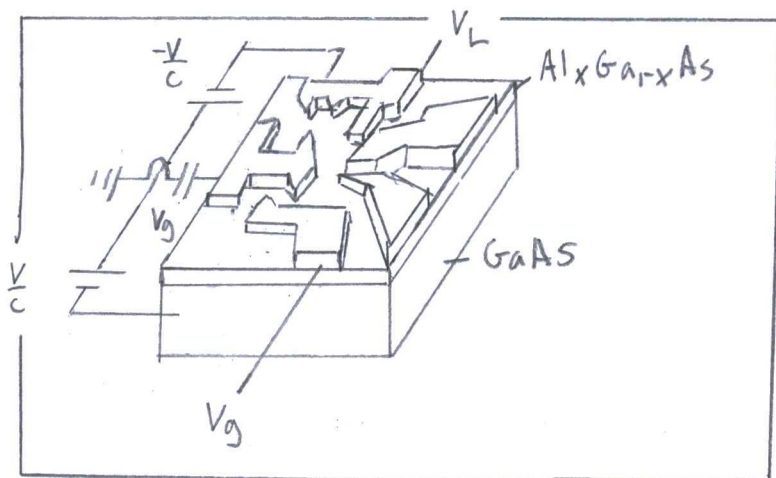
Hyperbolic identity:

$$\cosh(x) = \frac{e^x + e^{-x}}{2}$$

Greens Function:

$$\begin{aligned} G_{n\alpha} &= T \sum F(\tau) G(\tau) \\ &= \frac{T}{2\pi} \int d\omega \frac{2E_c}{(E_c^2 + \omega_m^2)} \frac{1}{(i\omega - \epsilon + \mu)} \\ &= \frac{1}{i\omega_n - E_c \operatorname{sgn}(\xi_n) - \xi_n} \end{aligned}$$

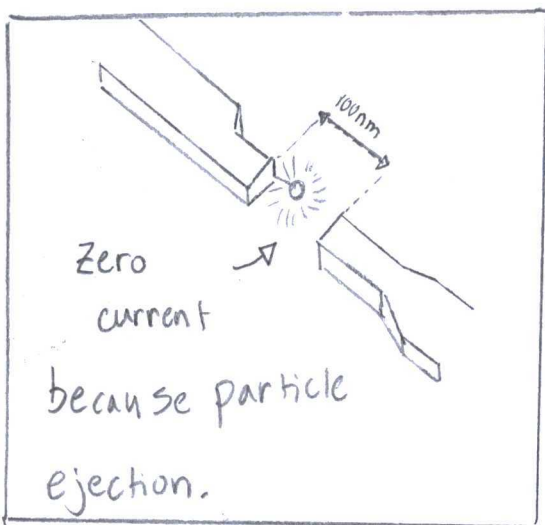
Problem 5.6.4 - Coulomb Blockade §



People scan voltages for "zero-points" in current near sub-kelvin temperatures.*

a) (5.61) "Action Potential"

$$S[\Psi] = \int_0^\beta d\tau \left(\sum_{\alpha} \bar{\Psi}_{\alpha} (\partial_{\tau} + \epsilon_{\alpha} - \mu) \Psi_{\alpha} + E_c \left(\sum_{\alpha} \bar{\Psi}_{\alpha} \Psi_{\alpha} - N_0 \right)^2 \right)$$



(3.71) "Field Integral"

$$Z = \int D\Psi e^{-S[\Psi]} = \int D\Psi e^{-\int_0^\beta d\tau \left(\sum_{\alpha} \bar{\Psi}_{\alpha} (\partial_{\tau} + \epsilon_{\alpha} - \mu) \Psi_{\alpha} + E_c \left(\sum_{\alpha} \bar{\Psi}_{\alpha} \Psi_{\alpha} - N_0 \right)^2 \right)}$$

$$= \sqrt{\frac{1}{4\pi E_c}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} D\Psi D\phi e^{\int_0^\beta d\tau \left[\sum_{\alpha} \bar{\Psi}_{\alpha} (\partial_{\tau} + \epsilon_{\alpha} - \mu) \Psi_{\alpha} - \frac{\phi^2}{4E_c} + i(N_0 + \sum_{\alpha} \bar{\Psi}_{\alpha} \Psi_{\alpha}) \phi \right]}$$

"The coulomb blockade"

Hubbard - Stratonovich Identity:

$$e^{-\frac{a}{2}x^2} = \sqrt{\frac{1}{2\pi a}} \int_{-\infty}^{\infty} e^{-\frac{y^2}{2a} - ixy} dy$$

* Deposition, low temperature, microscope, and math.

(from problem) "Green Function"

$$\begin{aligned}
 G_\alpha(\tau) &= \frac{1}{Z} \int DV e^{-S[V]} \circ Z_n[V] \circ G_\alpha[V](\tau) \\
 &= \frac{1}{Z} \sqrt{\frac{1}{4\pi E_c}} \int D^4V e^{-\int_0^\beta d\tau [\sum_\alpha \bar{\psi}_\alpha (\partial_\tau + \epsilon_\alpha - \mu) - \frac{V^2}{4E_c} + i(N_0 V + \bar{\psi}_\alpha V \psi)]} \\
 &= \frac{1}{Z} \sqrt{\frac{1}{4\pi E_c}} \int DV e^{-\int_0^\beta d\tau (\frac{V^2}{4E_c} - i(N_0 V + \bar{\psi}_\alpha V \psi))} \circ Z_\alpha[V] \circ G[V](\tau)
 \end{aligned}$$

b) Gauge Transformation:

(from problem) "Gauge Transformation"

$$A_\mu \longrightarrow A_\mu + \partial_\mu \Upsilon$$

where Υ is an arbitrary function.

$$\text{So, } V = V(\tau) + \hat{V}$$

$$\begin{aligned}
 G^{(\kappa)} &= \frac{1}{Z} \sqrt{\frac{1}{4\pi E_c}} \int DV \circ e^{-S[V]} \circ Z_\alpha[V] \circ G_\alpha[V](\tau) \\
 &= \frac{1}{Z} \sqrt{\frac{1}{4\pi E_c}} \int DV e^{-S[V]} \circ e^{i \int_0^\beta d\tau (V(\tau) + \hat{V})} \circ Z_\alpha[V] G_\alpha[V](\tau) \\
 &= \prod_m \frac{\int dV e^{i \int_0^\beta d\tau (V(\tau) + \hat{V})}}{\beta Z} \sqrt{\frac{1}{4\pi E_c}} \int DV \circ e^{-S[V]} \circ Z_\alpha[V] G_\alpha[V](\tau)
 \end{aligned}$$

Partial Fractions for Complex Values:

$$\frac{1}{2\pi} \int_0^{\infty} \frac{dw Z E_c}{(E^2 + w^2)(iw - b)}$$

$$b = \epsilon - \mu$$

$$= -\frac{iET}{\pi} \int \frac{dw}{(E^2 + w^2)(iw - b)}$$

$$= -\frac{iET}{\pi} \left[\int \frac{A}{(w + ib)} dw + \int \frac{Bw + C}{(E^2 + B^2)} dw \right]$$

Unknown A:

$$1 = A(w^2 + E^2) + (Bw + C)(w + ib)$$

$$= A(w^2 + E^2) \quad @ w = -ib$$

$$A = \frac{1}{E^2 - b^2}$$

Unknown C:

$$1 = A(w^2 + E^2) + (Bw + C)(w + ib)$$

$$-\frac{(b-w)(b+w)}{E^2 - b^2} = (Bw + C)(w + ib)$$

$$C = \frac{-ib}{E^2 - b^2} \quad @ w = 0$$

Unknown B:

$$1 = A(w^2 + E^2) + (Bw + C)(w + ib)$$

$$-\frac{(b-w)(b+w)}{E^2 - b^2} = (Bw + C)(w + ib)$$

$$\beta = -\frac{1}{E^2 - b^2}$$

$$= -\frac{iE}{\pi} \frac{T}{E^2 - b^2} \int_0^\infty \frac{1}{w + ib} dw - \int_0^\infty \frac{w - ib}{(E^2 + w^2)} dw$$

$$= \frac{2iET \ln(b) + \pi T b - 2iT E \ln(E) - \pi T(E)}{2\pi(E^2 - b^2)}$$

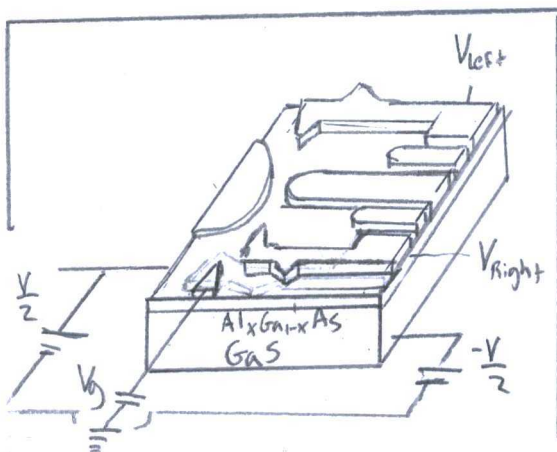
$$\text{Re}(t) = \frac{T}{2(E - \epsilon + \mu)}$$

e) IF $V(\epsilon) = E - \epsilon + \mu$
 > 0

then $E < \epsilon - \mu$

When particle energy is greater than chemical potential.

5.6.5 - Action of a Tunnel Junction:



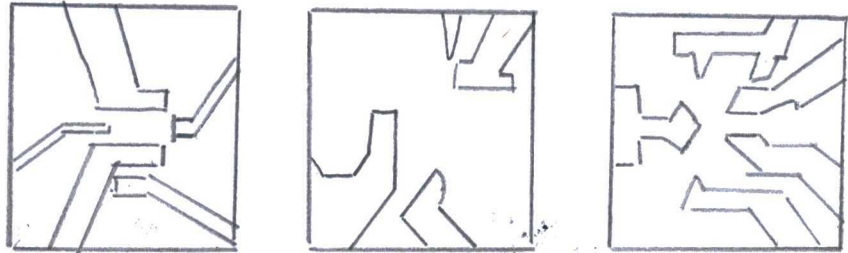
Cite: Petta, J. R.

"Coherent Manipulation of Coupled Electronic Spins in Semiconductors Quantum Dots.

Boston, Ma., (2005)

The paper scans voltages at each terminal for current at V_g .

Other shapes and cathodes:



... some cathodes failed, similar to Edison, Thomas.

a) (from problem) "Total action"

$$S[\psi_\alpha, \psi_a] = S_{\text{dot}}[\psi_\alpha] + S_{\text{lead}}[\psi_a] + S_T[\psi_\alpha, \psi_a]$$

(from problem) "Action lead"

$$S_{\text{lead}}[\bar{\psi}_a, \psi_a] = \sum_a \int_0^B d\tau \bar{\psi}_a (\partial_\tau + \epsilon_a - \mu) \psi_a$$

(from problem) "Action coupling between dot and lead"

$$S_T[\psi_\alpha, \bar{\psi}_\alpha, \psi_a, \bar{\psi}_a] = \sum_{\alpha a} \int d\tau \bar{\psi}_\alpha T_{\alpha a} \psi_a + h.c.$$

(5.61) "Action" integral

$$S[\psi] = \int_0^{\beta} d\tau \left(\sum_{\alpha} \bar{\psi}_{\alpha} (\partial_{\tau} + \epsilon_{\alpha} - \mu) \psi_{\alpha} + E_c (\sum_{\alpha} \bar{\psi}_{\alpha} \psi_{\alpha} - N_0)^2 \right)$$

(3.71) "Field Integral"

$$\begin{aligned} Z &= \int D\psi e^{-S[\psi]} \\ &= \int D\psi e^{-(S_{\text{dot}}[\psi] + S_{\text{lead}}[\psi] + S_T[\psi_{\alpha}, \psi_a])} \\ &= \int_{-\infty}^{\infty} D\psi e^{-\int_0^{\beta} d\tau \sum_{\alpha} \bar{\psi}_{\alpha} (\partial_{\tau} + \epsilon_{\alpha} - \mu) \psi_{\alpha} + \frac{V^2}{4} - i(N_0 V) + iV} \\ &= \int_{-\infty}^{\infty} D\psi DVe^{-\sum_{\alpha} \int \bar{\psi}_{\alpha} (\partial_{\tau} + \epsilon_{\alpha} - \mu) \psi_{\alpha} - \sum_{\alpha} \bar{\psi}_{\alpha} T \psi_{\alpha}} \end{aligned}$$

$$= \frac{\pi}{4\beta^{3/2}} \int_{-\infty}^{\infty} D\psi DV e^{-S_{\text{eff}}}$$

Where $S_{\text{eff}}[V] = S_c[V] = \frac{1}{2} \ln \begin{pmatrix} \partial_{\tau} - \mu + \epsilon & T \\ T & \partial_{\tau} - \mu + \epsilon \end{pmatrix}$

$$S_c[V] = \int d\tau \frac{V^2}{4} - i(N_0 + \bar{\psi}_{\alpha} \psi_{\alpha})V$$

... the next steps to the book's

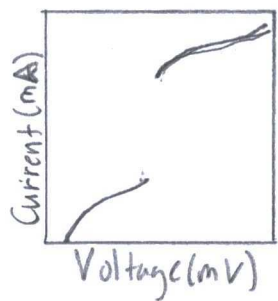
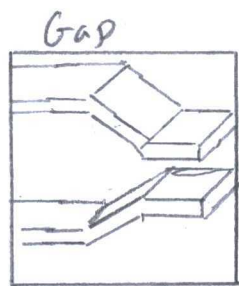
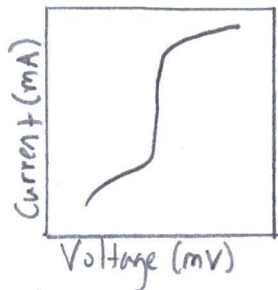
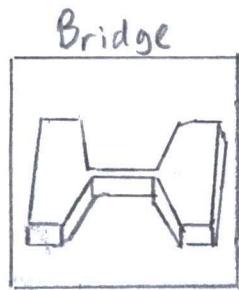
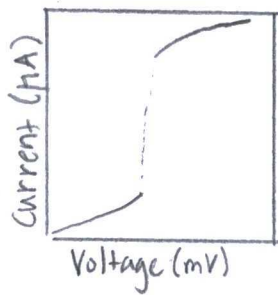
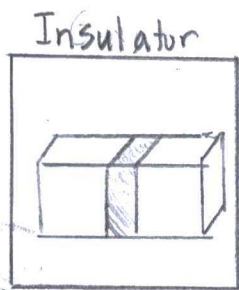
goal stopped because...

$$\sin(x) = x - x^3/3 + \dots$$

$$\sin^2(x) = x^2 + O(x^4)$$

$$\sin^2(T^2) = T^2$$

Problem 5.6.6 - Josephson Junction:



a) (from problem) "Action Quantum Dot"

$$S^i[\psi_\alpha^i, \phi_i] = \int_0^{\beta} d\tau \sum_{\alpha} \bar{\psi}_{\alpha}^i (\partial_{\tau} + \xi_{\alpha} \sigma_3 + e \Delta \sigma_1) \psi_{\alpha}^i$$

(from problem) "Capacitive Interaction"

$$S_{int} = \frac{E_c}{4} \int d\tau (\hat{N}_1 - \hat{N}_2)^2$$

$$\text{where } \hat{N}_i = \sum_{\alpha} \bar{\psi}_{\alpha}^i \sigma_3 \psi_{\alpha}^i$$

(from problem) "Two dots"

$$S_T[\psi_{\alpha}] = \sum_{\alpha\beta} \int d\tau \bar{\psi}_{\alpha}' (T_{\alpha\beta} \sigma_3) \psi_{\beta}^2 + h.c.$$

(3.71) "Field Integral"

$$Z = \int d\psi e^{-S[\psi]}$$

$$= \int d\psi e^{-S^i[\psi_{\alpha}, \phi_i] - S_{int} - S_T[\psi_{\alpha}]}$$

$$= \sqrt{\frac{1}{4\pi E_c}} \int D\psi D\phi e^{-\int_0^{\beta} d\tau \sum_{\alpha} \bar{\psi}_{\alpha}^i (\partial_{\tau} + \xi_{\alpha} \sigma_3 + e \Delta \sigma_1) \psi_{\alpha}^i}$$

$$- \frac{V^2}{4E_c} + iV(N_1 - N_2)^2$$

$$+ \sum_{\alpha\beta} \bar{\psi}_{\alpha}' (T_{\alpha\beta}) \psi_{\beta}^2$$

$$= \sqrt{\frac{1}{4\pi E}} \sqrt{\frac{\pi}{\beta(2\tau + \xi_1 \sigma_3 + e^{i\phi(\tau)\sigma_3} \Delta\sigma_1) + T}} \sqrt{\frac{\pi}{\beta T \sigma_3}} \times$$

$$\int d\tau e^{-\int_0^\beta d\tau \frac{V^2}{4E_0}} \quad \dots \text{When } N_1 = N_2$$

$$= \frac{\pi}{2 \sqrt{\beta^3 E}} \int d\tau e^{-\int_0^\beta d\tau \frac{V^2}{4E_0}} \cdot e^{\ln(2\tau + \xi_1 \sigma_3 + e^{i\phi(\tau)\sigma_3} \Delta\sigma_1 + T)}$$

$$= \frac{\pi}{2 \sqrt{\beta^3 E}} \int d\tau e^{-\int_0^\beta d\tau \frac{V^2}{4E_0}} \cdot e^{-\frac{1}{2} \ln \left(\frac{2\tau + \xi_1 \sigma_3 + e^{i\phi(\tau)\sigma_3} \Delta\sigma_1}{T} \cdot \frac{T}{2\tau + \xi_1 \sigma_3 + e^{i\phi(\tau)\sigma_3} \Delta\sigma_1} \right)}$$

$$= \frac{\pi}{2 \sqrt{\beta^3 E}} \int d\tau e^{-S_{\text{eff}}[V]}$$

$$\text{Where } S_{\text{eff}}[V] = S_c[V] - \ln \begin{pmatrix} 2\tau + \xi_1 \sigma_3 + e^{i\phi(\tau)\sigma_3} \Delta\sigma_1 & T e^{(\phi_1 - \phi_2)\sigma_3} \\ T e^{(\phi_1 - \phi_2)\sigma_3} & 2\tau + \xi_1 \sigma_3 + e^{i\phi(\tau)\sigma_3} \Delta\sigma_1 \end{pmatrix}$$

$$\text{When } S_c[V] = \frac{1}{4E_0} \int d\tau V^2$$

$$b) \begin{pmatrix} 2\tau + \xi_1 \sigma_3 + e^{i\phi(\tau)\sigma_3} \Delta\sigma_1 & T e^{(\phi_1 - \phi_2)\sigma_3} \\ T e^{(\phi_1 - \phi_2)\sigma_3} & 2\tau + \xi_1 \sigma_3 + e^{i\phi(\tau)\sigma_3} \Delta\sigma_1 \end{pmatrix}$$

$$\underline{\text{ansatz}} = \begin{pmatrix} G^{-1}(V, \phi) & T \\ T & G^{-1}(V, \phi) \end{pmatrix}$$

$$S_{\text{eff}}[V, \phi] = \frac{1}{G(V, \phi) G(V, \phi)} - T^2$$

ln Series
 $\ln(1+x) = x - O(x^2)$

$$S_{\text{eff}}[V, \phi] = S_c[V] - \frac{1}{2} \ln \left(\frac{1}{G(V, \phi) G(V, \phi)} - T^2 \right)$$

$$\simeq S_c[V] - \frac{1}{2} \left(\frac{1}{G(V, \phi) G(V, \phi)} - T^2 - 1 \right)$$

$$\simeq S_c[V] - T^2 \frac{G(V, \phi) G(V, \phi)}{G(V, \phi) G(V, \phi)} - \dots$$

~~~~~

$$S_{\text{tan}}[\phi]$$

$$S_{\text{tan}}[\phi] = \frac{T^2 G(V, \phi) G(V, \phi)}{G(V, \phi) G(V, \phi)}$$

where  $G(V, \phi) = (i\omega_n - \xi\sigma_3 - \Delta\sigma_1)^{-1}$

c) If  $S_{\text{tan}}[\phi] = |T|^2 \sum G_1(e^{i\sigma_3\phi}) G_{2, \alpha, \omega_n + \omega_m}(e^{-i\sigma_3\phi})$

where  $G_{i, \omega_n} = (i\omega_n - \xi\sigma_3 - \Delta\sigma_1)^{-1}$

$$S_{\text{tan}}[\phi] = |T|^2 \text{tr}(\sigma_1(e^{i\sigma_3\phi}) \sigma_1(e^{-i\sigma_3\phi}))$$

so somewhat idiomatic because  $e^{i\sigma_3\phi} \circ e^{-i\sigma_3\phi}$

is one.

$$= |T|^2 (\pi\rho)^2 \Delta \int d\tau \cos(2\phi(\tau))$$



$$\text{Where } \gamma = |T|^2 (\pi \rho)^2 \Delta$$

If  $V = \phi + V_0$ , then,

$$\begin{aligned} S[\phi] &= \int_0^\beta d\tau \frac{V^2}{4E_c} + \gamma \int_0^\beta d\tau \cos(2\phi(\tau)) \\ &= \frac{1}{4E_c} \int_0^\beta d\tau \dot{\phi}^2 + \gamma \int_0^\beta d\tau \cos(2\phi(\tau)) \end{aligned}$$

d) (from problem) "Action of the Josephson Junction"

$$S[\phi] = \frac{1}{4E_c} \int d\tau \dot{\phi}^2 + \gamma \int d\tau \cos(2\phi(\tau)) + T S_{\text{diss}}[\phi]$$

$$\text{If } \dot{\phi} = i 2E_c N \quad \text{and} \quad \dot{N} = 2i \gamma \sin(2\phi)$$

$$S[\phi] = E_c \int d\tau N^2 + \gamma \int d\tau \cos(2\phi(\tau)) + i \int d\tau N \dot{\phi}$$

$$\text{When } I = i 2E_c N$$

$$= -2\gamma \sin(2\phi)$$

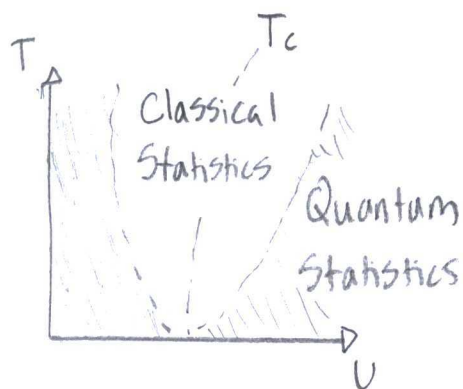
$$\text{Since } S_{\text{tan}}[\phi] = \gamma \int_0^\beta d\tau \cos(2\phi(\tau))$$

$$I = 2E_c^2 S_{\text{tan}}[\phi]$$

$$= -2\gamma \sin(2\phi)$$



# Problem 5.6.7 - Metallic Magnetism:



Cite: Hertz, J.A.

"Quantum critical phenomena"

Chicago, Illinois (1975)

a) ( 3.71 ) "Field Integral"

$$Z = \int D\psi \exp \left\{ - \int_0^\beta d\tau \left( \sum_p \bar{\psi}_p (2\tau + \xi_p) \psi_p - \frac{U}{4} \sum_i (\psi_i \sigma_3 \psi_i)^2 \right) \right\}$$

$$= \int Dm D\psi \exp \left\{ - \int_0^\beta d\tau \left( \frac{U}{4} \sum_i m_i^2 + \sum_p \bar{\psi}_p (2\tau + \xi_p) \psi_p + i \frac{U}{2} \sum_i \psi_i \sigma_3 m \psi_i \right) \right\}$$

$$= \sqrt{\pi} \cdot \exp \left( \frac{-U}{4} \int_0^\beta d\tau \sum_i m_i^2(\tau) + \ln \left( 1 - \frac{iU}{2} \sigma_3 m \epsilon \right) \right)$$

... when  $2\tau + \xi = 1$ ,  $\sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

$$b) S[m] = \underbrace{-\frac{U}{4} \int_0^\beta d\tau \sum_i m_i^2(\tau)}_{S^{(4)}[m]} + \underbrace{\ln \left( 1 - \frac{iU}{2} \sigma_3 m \epsilon \right)}_{S^{(2)}[m]}$$